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Beyond zero-sum: game theory and national forest management[☆]

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Abstract

Following two decades of bitter struggle, the current debate over management of the U.S. national forests is characterized by animosity, controversy, and seemingly intractable gridlock. This view has led some to characterize the current management process a situation of pure conflict or a zero-sum game. In this article, we develop a game-theoretic model based on an argument that national forest policymaking is a game between the U.S. National Forest Service and Environmentalists and that this game is not zero-sum. The model is used to account for outcomes associated with contemporary management policy and to examine some recent changes to the game's rules and how this may affect outcomes. The analysis shows that some changes will have little to no effect on outcomes, while others have a significant potential to do so.

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1. Introduction

The summer of 2003 marked the third consecutive year that large fires ravaged the forests of the Western United States. These fires, which collectively burned millions of acres and dozens of homes, have brought into sharp focus a long-running debate about forest management policies and led to widespread demands by the public and political actors that something be done to prevent a reoccurrence. As is often the case “what to do” is easy to identify, “how to do it” is much more challenging.

A wide-range of commentators, critics, and participants contend the management of the national forests has become increasingly problematic. The seemingly irreconcilable demands

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of competing interests and a forest management decision-making process characterized by conflict, litigation, and proactive, extensive, participation by private actors inevitably produces inaction that forestalls efforts to prevent the kinds of catastrophic events mentioned above (USDA, 2002a). In addition, it is argued this increasingly dysfunctional process, and the inaction that results, is primarily a product of the contentious relationship that exists between the timber industry and environmental organizations, who battle over forest policy while locked in a zero-sum game (Bosworth, 2002).

This zero-sum characterization of the process is intriguing and raises several interesting questions. To what extent is the current policymaking process best modeled as a “game” between the timber industry and the environmentalists? Is this game best modeled as zero-sum? Under what conditions does the outcome of this game lead to inaction and how might the game be changed? The purpose of this article is to shed light on these questions by developing a game-theoretic model to capture what is commonly understood as the essential “problem” in managing national forest lands.

We contend that among all the competing actors and different relationships involved in making national forest policy, the one that appears to be of greater importance is not the game played by *timber* and the *environmentalists* but the game that is played by the U.S. *Forest Service* (USFS) and the *environmentalists*. We develop a model based on the assumption that the USFS and the environmentalists are the key actors and it reveals that once the complete game has been specified, it is not zero-sum. Further, we argue that it is unnecessary to develop a zero-sum game in order to explain the outcomes associated with this aspect of the current national forest policymaking game.

These outcomes are a tremendous source of frustration for the USFS and have led many within and outside the agency to advance a number of changes that are intended to alter the game’s outcomes (USDA, 2002a). Among the changes recently enacted, we analyze three in order to determine whether, or to what extent, they will alter the outcomes of the current game. Importantly, we find that some changes will have little to no effect on the present game, while others have the potential to transform fundamentally the outcomes generated by the forest management policymaking process.

In the first section, we discuss the origins of the current conflict among the principal actors and the generic process of forest management policymaking. These observations are used to formulate the game-theoretic model developed in the second section. In the third section, we analyze the potential effects of some recent changes to forest management policymaking. In our concluding remarks, we suggest that the problematic nature of national forest policymaking is not simply a case of institutional failure. Further, we contend that some of the recent changes in the decision-making processes, which alter the forest management game in the short-term, may actually undermine the wide-ranging efforts to promote long-term consensus and collaborative approaches to national forest management.

2. Origins of a misconception

The 1960s saw the American public showing a much greater interest in the management of the national forests. A developing environmental awareness and a growing concern over the

large-scale comodification of the nation's forest resources led to increased public scrutiny and a number of legislated changes to forest management policies. Beginning with the Multiple Use Sustained Yield Act (1960), the goal of these changes was to promote non-timber values and increase pressure on the USFS to rethink its emphasis on large-scale timber production. The Sustained Yield Act was followed by the Wilderness Act (1964) and the National Forest Management Act (1976). In addition to these efforts to recast the scope and substance of national forest management priorities, the principal environmental protection legislation of the era (National Environmental Policy Act, 1969; Endangered Species Act, 1973) also furthered change in how the national forests were managed and what they were managed for.

The clash of ideas (ecological vs. extractive) and interests (timber industry and its supporters vs. environmental community) that followed grew in scope and substance in the late 1970s. It eventually culminated in the “war of the woods” that erupted in the late 1980s over the listing of the spotted owl as an endangered species and the closure of some national forest lands to timber harvesting (Dietrich, 1992; Yaffee, 1994). This conflict engendered a decade of suspicion, bitterness, and ritualized combat between “timber” and “environmentalist.” The result is a deeply polarized policymaking environment where war is the dominant metaphor and the language used to describe the relationship includes terms like “enemy,” “capitulation,” “victory,” and “defeat” (Salazar & Alper, 2000). In a recent assessment, Limerick suggests the conflict has the characteristics of the Cold War between the U.S. and the U.S.S.R. and notes that many participants, in this example the timber industry, continue to view the relationship in these terms. “The environmentalists are dedicated and determined to win the war. We should be, too” (Limerick, 2002, p. 46).

The use of the war metaphor serves to capture a generally held understanding of the character of the relationship between the timber industry and environmental organizations. The relationship has been further characterized by observers as a zero-sum game that leads inevitably to a dysfunctional policymaking process. “In the zero-sum game,” U.S. Forest Service Chief Bosworth observes, “folks measure their own success in terms of the misfortune of their adversaries” (2002, p. 3).

So if you're in the timber industry and you see something the environmentalists like, you automatically think it must be a bad idea, even if it's no skin off your nose. Or if you're an environmentalist and you see some folks getting some jobs from material we need to remove, you automatically think we must be in bed with the timber industry. Your focus is not on the land and what it needs. Instead, your focus is on how well your adversaries are doing, because if they seem to be winning, you must be losing. That's the zero-sum game (Bosworth, 2002, pp. 3–4).

Despite the engaging, intuitive, nature of this explanation, we argue in the following sections that both the war and the zero-sum game metaphors are misleading.

3. Understanding the game context

The creation of forest management policy is a complicated process that involves several different public and private actors. Among the most prominent are the timber and other extractive industries, environmental organizations, and recreational users. On any given issue, the

aims of these groups might overlap. But, more often than not, their interests may diverge and result in very different prescriptions regarding the management of forest lands. To promote and protect their interests, these actors may choose to press their claims through Congress or make demands of the Forest Service.

The foregoing brief sketch suggests that there are a variety of different “games” involved in making forest management policy. Sometimes private groups will initiate a game with the USFS by making a proposal that it can accept or reject. Other times, private interests will find themselves in a position where they have to react to a proposal made by the USFS. Aside from the issue of which actor commences the process, the outcomes of these games are also a function of the types of actors that select themselves into the process. Clearly, not all environmental groups share the same “worldview.” Nor, do actors always share the same preferences over policy within the extractive community or even within the USFS. Put differently, some environmental groups (and the timber industry or recreational users) may be prepared to fight the USFS irrespective of the agency’s proposal, while other groups will not block the agency either because conflict is too costly or because they have a favorable view of a proposed action. Similarly, there are times when the USFS may go to great lengths to defend its proposals (or to block private groups) and other times when inaction is perceived as a better course of action.

The USFS is often viewed merely as a conduit for timber industry preferences. This characterization is inaccurate given the USFS has historically had its own set of preferences that at times have differed sharply from those of the timber industry (Hirt, 1994; Yaffee, 1994). Environmental organizations obviously appeal and file legal challenges to contest commodity production decisions (logging). Yet, the USFS is also challenged by environmental organizations over decisions that provide no direct benefits to the timber industry. Decisions on recreational access issues (snowmobiles and all-terrain vehicles), ski resort expansion or construction, and restoration activities that may damage salmon habitat have all been subject to challenge from environmentalists.

Moreover, while timber production from national forest lands is of vital economic importance to a number of small communities in the west, the industry, particularly the larger corporations, has been reducing its operations in western national forests and has shifted much of its attention to other domestic and international sources. As a result, in the west, where the management of national forests is most contested, the timber industry has in most respects been replaced by the environmental community as the principal external actor shaping national forest policymaking. Thus, the attention given the relationship between timber and environmentalists is misdirected. Instead, it is the relationship between the USFS and the environmentalists that is of greater importance and a game-theoretic model that explores this relationship has much greater explanatory utility. We now turn to a specification of that game.

4. The game

It is important to note there is an interesting “pre-game” dynamic that ensues before the game can begin. This pre-game action involves regulatory agencies (Environmental Protection Agency, U.S. National Marine Fisheries Service, and U.S. Fish and Wildlife Service) that

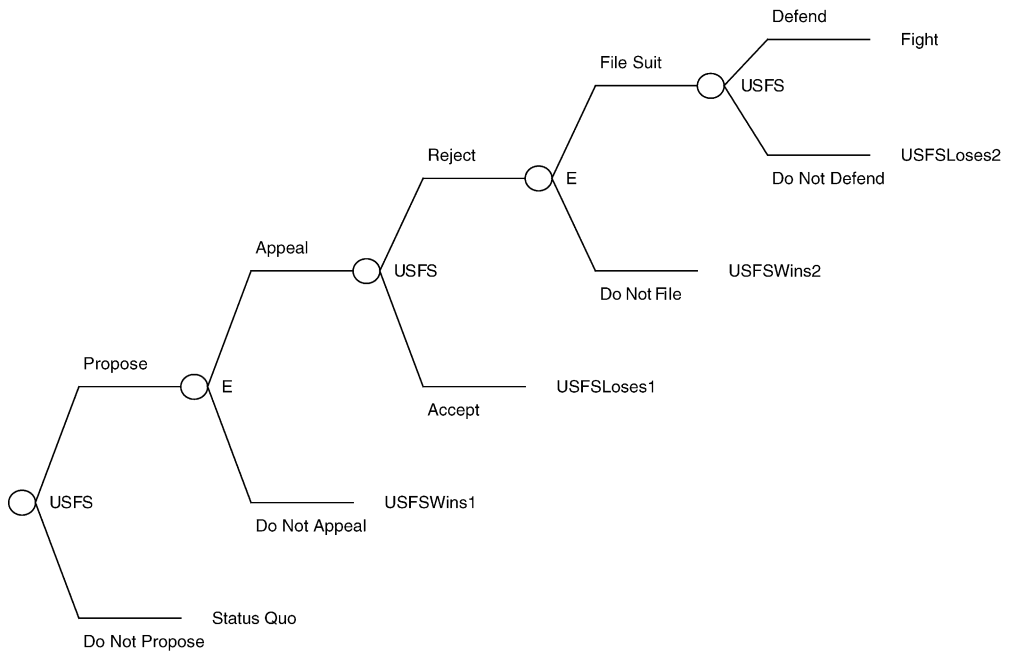


Fig. 1. USFS and environmentalists' game tree.

scrutinize and must ultimately approve many USFS decisions before the game even starts (USDA, 2002a). Despite the importance of this pre-game dynamic, and the reality that in many instances it often occurs simultaneously with the other game, we assume this process has resolved itself so that we can move forward to analyze the second stage of the game involving the USFS and the environmentalists.

The game is shown in Fig. 1. There are two rational actors in the game. We assume that the USFS begins the game and can choose between one of two acts.¹ The USFS can either make a proposal to initiate an action or choose not to make a proposal. There are several actions the USFS can propose, including timber sales, salvaging timber, removing dead and dying trees, reducing fire-prone underbrush, expanding recreational opportunities, and habitat restoration. If the USFS chooses not to make a proposal, the game ends and the outcome is designated as the Status Quo. In our game, the Status Quo prevails whenever the USFS chooses not to make a proposal. In other words, a Status Quo outcome means that nothing occurs or that national forest lands remain unchanged. The game ends without environmentalists (E) having to move. In contrast, the USFS frequently describes the Status Quo not as an outcome produced by its own initial inaction, but as an outcome that invariably leads to inaction even though a proposal was made initially (USDA, 2002a). Despite the different emphases, the Status Quo means the same thing conceptually—inaction.

If the USFS makes a proposal, the game continues and the E get an opportunity to act. E can either file an administrative appeal or choose not to appeal the Forest Service's proposal. If E does not appeal, the game ends and the outcome is denoted USFSWins1 (E Loses). If E appeals, the USFS has two options—it can accept or reject the appeal. If the appeal is accepted, the game ends and the outcome is USFSLoses1 (E Wins). If the appeal is rejected, E has one

more opportunity to move and chooses between filing and not filing suit against the USFS. If E does not file, the outcome is considered USFSWins2 (E Loses). If E files suit, the USFS must choose between defending and not defending itself in Federal Court. If the USFS fails to defend itself the game ends and outcome is designated USFSLoses2 (E Wins). If the USFS defends itself, the outcome is designated as a Fight.²

The next step is to specify the players' preferences over the six outcomes in the game. Starting with the environmentalists, we assume that E most prefers the Status Quo > to USFSLoses1 > to USFSLoses2 > to Fight > to USFSWins1 > to USFSWins2. A few words justifying this set of preferences are in order. First, note that there are two outcomes where E wins (USFSLoses1 and USFSLoses2) and two outcomes where E loses (USFSWins1 and USFSWins2). Here we assume that if a player is going to win (lose), it prefers to win (lose) earlier, rather than later, in the game. The reason is that even though the gains or losses associated with each of these win or lose outcomes are the same, the costs to achieve those outcomes are not. Outcomes that occur later in the game are more costly in terms of time, money, energy, etc., and are less preferred to wins or losses that occur earlier in the game. This same assumption will also apply to the Forest Service's preference ordering. We also assume that E prefers a Fight to both outcomes that involve giving in and losing to the USFS. An E with the foregoing preferences is considered to be a resolved actor.³

Now consider the preferences of the USFS. We assume the USFS may be one of two types—a resolved USFS or an irresolute USFS. The preference ordering associated with a resolved USFS is USFSWins1 > to USFSWins2 > to Fight > to Status Quo > to USFSLoses1 > to USFSLoses2. The only difference between a resolute and an irresolute USFS is that the latter reverses the preference ordering of the Status Quo and Fight outcomes, i.e., the Status Quo is preferred to a Fight. Resolved USFS types prefer to take the gamble on the payoff that results from a Fight over the alternative of failing to make a proposal. Irresolute USFS types prefer to do nothing rather than to take an action that leads to a Fight. Like E, both USFS types most prefer to win, while their worst outcome is losing to the other side.

Like E, we assume that both USFS types also prefer the Status Quo to losing. The logic is as follows. The outcomes, USFSLoses1 and USFSLoses2, occur when the USFS gives in to E by either accepting its appeal or failing to defend itself in a lawsuit, respectively. Either way, the USFS is blocked from implementing its proposal, inaction occurs, and forest lands remains unchanged. This is precisely the outcome that would have occurred had the USFS never made a proposal in the first place. Put differently, since the Status Quo is also a "USFS loses" outcome, and using the previous assumption that actors prefer the least costly outcome (when the benefit and/or loss is identical), it follows that both USFS types prefer the Status Quo > to USFSLoses1 > to USFSLoses2.

4.1. *Not zero-sum*

In our description of the game's outcomes, we consider a win for the USFS as a loss for E and a loss for the USFS as a win for E. There is little debate that the USFS perceives itself as having lost whenever it is blocked from implementing a proposal (Bosworth, 2002; USDA, 2002a). At issue are the reasons why the implementation of a USFS proposal could plausibly be considered a loss for E. We argued previously that the historical animosity between

the timber industry and the environmentalists continues to shape the latter's perception that the USFS is merely a conduit for timber industry preferences. This misperception structures E's understanding of the USFS's actions. In addition, environmental organizations may lack confidence in the USFS's willingness or ability (e.g., funding limitations) to keep its promises to implement certain decisions. Given the controversy over the implementation of the Quincy Library agreement or, more recently, rehabilitation efforts in Montana's fire-ravaged Bitterroot Valley, it is reasonable to conclude that E may consider USFS inaction as a better outcome than "trusting" the Forest Service to implement properly a proposal (Devlin, 2003). In these circumstances, a win for the USFS is considered a loss for E.

This characterization of the win–lose outcomes is consistent with the intuition behind zero-sum games. However, given our assumptions regarding the actors' preferences, the game between the USFS and E cannot be characterized properly as zero-sum. The technical definition of a zero-sum game is that all the payoffs sum to zero (Binmore, 1982). That is, the gain for one actor translates into an equivalent loss for the other. Even though the game at hand contains outcomes that are consistent with a zero-sum metaphor (i.e., USFS wins means E loses and vice versa) it is implausible to assume that the players' payoffs for either the Fight or Status Quo sum to zero. The reason for this is that although the Status Quo is the best outcome for E, it is not the worst for the USFS. The zero-sum definition also fails to hold for the Fight outcome.

We solve for the game's equilibria using backward induction under the condition of complete information. When the USFS is resolved, the equilibrium outcome is a Fight. When the USFS is irresolute, the equilibrium outcome is the Status Quo.⁴ It is interesting to note that many in the USFS contend they "cannot win" given the structure of the current game (USDA, 2002a). The derivation of the game's two possible equilibrium outcomes confirms this is indeed the case, although it is important to note that it is neither necessary nor sufficient to describe the game in zero-sum terms to account for these outcomes. Our game generates similar results.

Although the USFS can win theoretically at two different points in the game, neither of these outcomes will ever occur in equilibrium. The reason is that given E's preferences, E's optimal strategy is to file an appeal. If that appeal is rejected, E files suit. Since E prefers a Fight to the outcome it gets by failing to appeal (USFSWins1) and to the outcome when it does not file suit (USFSWins2), E will choose the act that blocks the Forest Service. In other words, E controls, through its decision, whether the USFS wins or not. Herein lies the frustration on the part of the USFS. It knows with certainty that E will try to block its proposals and will be undeterred by a resolved USFS that will defend its decisions in court. Thus, almost any proposal the USFS makes will lead to a Fight. The issue simply becomes a question of whether the USFS wants a Fight or would rather maintain the Status Quo.

5. Game change

The foregoing analysis strongly suggests the USFS will be able to win only when fundamental aspects of the current game are changed. Indeed, many both inside and outside the USFS have called for "game change" in order to restore the balance of competing interests (USDA, 2002a). There are two very different views on how best to achieve game change. They share, however, a common goal of constructing a win outcome for the USFS. The first view calls upon

the environmentalists to change their preferences via a reinterpretation of the outcome USFS wins (Bosworth, 2002). The second view takes the preference configuration of both players as given, but calls for legislation and administrative initiatives that will construct a USFS win by altering the rules of the current decision-making process. Each of these solutions is examined in turn.

5.1. *Game change via a new preference ordering*

Presently, a resolute E views a win for the USFS as its own loss. Instead of viewing “the USFS makes a proposal and E does not appeal” as a win–lose outcome, some argue that E should view the outcome as win–win, especially given that in theory the two actors share the same fundamental goal of maintaining a healthy ecosystem for current and future generations (Bosworth, 2002).

At issue, is the theoretical effect of this change on the game. If E were to reinterpret the meaning of the game’s outcomes along the foregoing lines, the utility associated with each of the outcomes would change and ultimately so would E’s preference ordering. Put differently, blocking a USFS proposal through the appeals process or through litigation would become less preferred than allowing the USFS to proceed with an action. Thus, the plea to reinterpret outcomes is really a plea for E to adopt the same preference ordering as the USFS: $USFSWins1 > to USFSWins2 > to [Fight > to Status Quo] \text{ or } [Status Quo > to Fight] > to USFSLoses1 > to USFSLoses2$. It does not matter whether the USFS is resolute or irresolute. E never appeals and the equilibrium outcome is $USFSWins1$. The implication, of course, is that since the preferences of the two actors are identical all conflict is eliminated and the game of forest management becomes a trivial matter.

5.2. *Game change via legislative or administrative initiatives*

Over the last decade, Congress has considered a number of proposals to alter the rules governing management of the national forests. These proposals generally included requirements to reduce the threat of wildfires (more harvesting of timber in certain areas and removal of fire-prone underbrush) and advocated amending the USFS’s decision-making process to make it more flexible and responsive to evolving circumstances. These two priorities were at the heart of a number of recently enacted changes included in the Bush Administration’s *Healthy Forest Initiative*. We examine three specific changes in the forest management decision-making process to assess how these might change the game in favor of realizing a win for the USFS. As we will demonstrate, these changes have only a limited capacity to construct a win–win outcome for both sides.

First, since 1993, citizens have had the right to an administrative appeal of a USFS decision to proceed with a proposed project.⁵ In June 2003, the U.S. Department of Agriculture released new rules that establish specific cases where these decisions are not subject to administrative appeal.⁶ Excluded from appeal are decisions that approve projects to reduce fire risk and that are less than 1,000 acres, or controlled burns of less than 4,500 acres.⁷

The elimination of the right to appeal certain decisions removes one of the options available to environmentalists. What remains is, the file-do not file suit sub-game and this reduces the

number of outcomes from six to four. We assume E prefers the Status Quo > to USFSLoses2 > to Fight > to USFSWins2. A resolved USFS prefers USFSWins2 > to Fight > to Status Quo > to USFSLoses2, while an irresolute USFS prefers USFSWins2 > to Status Quo > to Fight > to USFSLoses2.

If we hold all other changes constant and analyze just the effects of eliminating the administrative appeals option, the outcomes in the new game are identical to those of the previous game. The Status Quo holds when the USFS is irresolute and a Fight results whenever the USFS is resolved. Thus, eliminating the appeals option, by itself, does nothing to make a win more probable for the USFS. A USFS win outcome is still determined or controlled by E's decision over whether to file suit or not. And since E prefers both outcomes associated with filing a lawsuit (USFSLoses2 and Fight) to not filing suit (USFSWins2), E always litigates.

Many environmental groups contend this change will upend decades of environmental protection regulations and take away a meaningful public role in national forest management (Gerstenzang, Shogren, & Boxall, 2002). Our analysis, however, suggests that these concerns may be overstated in that abolishing the appeals process, by itself, will not change the current game in any meaningful way. If E retains the right to sue the USFS, E can still block the agency, albeit at a higher cost than through the appeals process.

The second important rule change we examine is included in the Healthy Forest Restoration Act of 2003.⁸ The most controversial elements of the Act are limits on the scope of judicial review of proposed USFS actions.⁹ Under the new guidelines, preliminary injunctions are limited to 60 days. Although injunctions are renewable, the parties involved in the dispute must now present updated information at the time a renewal is requested. The effects of these limits will be to reduce in most cases the time that elapses between a notice of an action and when a USFS project can begin, assuming the court rules in its favor. A favorable ruling is now made more probable via this Act since judges are required to take into account the short- and long-term effects of the proposed action compared to the effects that result from no action.¹⁰ This rule change will address a principal complaint of the USFS regarding its cumbersome and drawn-out decision-making procedures. This change, which ostensibly is for proposals to reduce hazardous fuels (i.e., fire-prone underbrush or dead trees), has been subject to sharp criticism by many environmental organizations. They contend that under the guise of reducing wildfire risk, the USFS will promote harvesting of old-growth timber and will engage in other actions that will harm ecological systems (Soraghan, 2003).¹¹

In terms of the game, this rule change affects the procedures and processes associated with a Fight outcome. First, this new rule has the potential to increase E's costs to litigate a dispute with the USFS. In the past, when E filed suit, generally on the grounds the USFS failed to follow ESA or NEPA procedural requirements, injunctions often remained in effect until the USFS could demonstrate procedural compliance or that the proposed action did not threaten endangered species. Thus, one of the principal complaints of the USFS is that even if successful in court, the time involved in arguing its case often delays action for so long conditions have changed to such an extent it must begin the process anew. Now with injunctions limited to 60 days, and given the requirements to obtain one or more renewed injunctions, if E wants to block the USFS it will have to devote more time and financial resources to generate new arguments or sustain existing ones. This requirement increases E's costs of a Fight without any guarantee that its efforts will be rewarded with another injunction. Second, the probability that

E loses in court (i.e., the USFS can proceed with a project) is also likely to increase. As noted, judges are now required to determine whether the harm associated with blocking USFS action is greater than the harm when the Forest Service is allowed to act. Since the former criterion was not considered formally in the past, the likelihood of inaction (USFS loses/E wins) was arguably higher.

At issue is the potential effect of the rule change on the game. On the one hand, there will be some environmental groups who perceive the stakes as high or even higher under the new rule (i.e., the potential harvesting of old-growth forests), whose resoluteness will remain intact, and will appeal and litigate even as costs rise and the probability of losing increases. Note that even though E's strategy remains the same, a resolute USFS is more likely to win given an accelerated time frame and new decision-making criteria. On the other hand, this new rule also carries the potential to transform some resolute E groups into irresolute ones. The difference between the two E types is their perception of the utility associated with a Fight.

Recall that a resolute E prefers a Fight to outcomes where the USFS wins—i.e., Status Quo > to USFSLoses1 > to USFSLoses2 > to Fight > to USFSWins1 > to USFSWins2. An irresolute E reverses the ordering of Fight and USFS wins: Status Quo > to USFSLoses1 > to USFSLoses2 > to USFSWins1 > to USFSWins2 > to Fight. An irresolute E still prefers to win but perceives a Fight as the worst possible outcome because the costs of a fight are deemed too prohibitive, the issue at stake is not particularly salient, and/or losing in court is very likely and not worth the cost.

Solving the game when E regains its full ability to appeal and E is irresolute reveals that the Status Quo continues to prevail when the USFS is also irresolute. However, when an irresolute E plays against a resolute USFS, the outcome is the Forest Service's most preferred, USFSWins1.¹² The reason is that since E realizes that it cannot win against a resolved USFS, it gives in immediately and does not appeal a proposal. Note that E's failure to appeal is not a function of E actually liking the proposal. E does not appeal because it can foresee that its worst outcome, a Fight, will result if it blocks the USFS. Unlike the elimination (or severe restriction) of the appeals process, this rule change has the potential to construct a win for the USFS, although it has limited capacity to construct a win-win situation for both sides.

The final rule change we examine is a case of the proverbial "stick and carrot." The USFS, in an effort to emphasize public comment and stakeholder consultation early in the decision-making process, has promoted extensively the value of public participation in the pre-decision stage (USDA, 2002b). The goal, obviously, is to involve private actors early in the process so their concerns are considered in any proposed action. To encourage participation, individuals or organizations that might want to participate in the administrative review process are now required to have submitted written comments on the proposed action during the public comment process. Further, in cases where the proposed action is for hazardous fuel reduction the parties may bring civil action only after a challenge in the administrative appeal process, and an issue may be considered in judicial proceedings only if it was raised in the administrative appeal process.¹³ The result of these changes will be to limit, if not in many cases eliminate, opportunities to challenge and perhaps block a proposed USFS action.

The effect of this rule change is straightforward. The original game we developed would cease to exist because E is largely prohibited from acting once a USFS proposal has been made. The new game would involve actors making choices and bargaining over the specific contents

of a proposal that eventually will be implemented by the USFS. The new game would be part of the pre-game policymaking process that we assumed away initially in order to analyze the effects of strategic interaction on the post-proposal stage.

To capture adequately the effects of this rule change, an entirely new model would have to be constructed and analyzed. Our purpose here, however, is more modest in that we offer non-formalized insights on what kind of game this actually might be and how this pre-decision game plausibly might unfold. First, it should be noted the USFS believes that the current game provides disincentives for E to cooperate initially, since E knows it can subsequently appeal or litigate (USDA, 2002a). By eliminating those acts, E will have greater incentives to cooperate with the USFS in developing policy choices given that E poses no credible threat to a proposed action once a decision to proceed has been made. If E loses its ability to influence the outcome once a proposal has been launched, then it will try to exert whatever influence remains toward shaping the contents of a plan. Whether this will result in a greater level of cooperation between the two actors remains an open question.

The issue here is not whether the two sides benefit from mutual cooperation. Rather, the main concern for E is whether or not the USFS can be trusted to implement what the two sides agreed upon. If the answer is no, the USFS cannot be trusted, then E might be wary of pursuing a cooperative strategy out of fear that it will be exploited during the implementation phase. On this view, E's worst outcome may result when both agree to cooperate, but the USFS actually defects from that agreement by permitting harvesting of old growth, new road construction, or simply failing to restore damaged habitat. Given that suspicion and mistrust plague the current relationship between E and the USFS, cooperation may be very difficult to establish in the short-run without the intervention of a neutral third party. Moreover, irrespective of the unfolding of this pre-game play, this rule change, almost by definition, constructs a win outcome for the USFS since it will be very difficult for E to block the USFS.

6. Conclusion

The USFS has been described as an example of institutional failure, unwilling or incapable of delivering on its promises to balance resource use equitably (Hirt, 1999; Nelson, 2000). Several factors explain this failure, including organizational bias, scientific disagreement within the agency, lack of Congressional funding, and the willingness of the USFS to sacrifice long-term environmental protection in favor of short-term economic, and politically expedient, goals (Nelson, 2000; Sedjo, 2000). Aware of this reputation, the USFS has tried to account for its problems by stressing the statutory, regulatory, and administrative considerations that shape national forest management (USDA, 2002a).

These considerations structure fundamentally the game of USFS decision making. In addition, the existing statutes and regulations endow other actors, governmental agencies and private groups (timber, environmentalists, etc.), with tools that allow them to scrutinize USFS decisions. This, in turn, facilitates a strategy of close oversight and active participation in the USFS decision-making process. It is not, however, simply the constraints imposed by existing statutes and regulations that produces the Forest Service's frustration with the game. Rather, it is how external actors view both the USFS and the decision-making process.

The USFS believes the long-term solution to its problematic forest management challenges is to build trust among its clients and to get those clients to view USFS actions as an outcome where everyone wins (Bosworth, 2002). However, the Forest Service's ability to promote and achieve collaborative solutions will be undercut severely by some of the recent changes to forest management decision making. One of the important findings of this study is that eliminating the administrative appeals process, all else being equal, will not alter the outcomes in the current forest management game. Conversely, redefining E's file suit act has the potential to undermine some E groups' willingness and/or ability to block proposals, thereby constructing the win desired by the USFS. We also argued that attempts to basically take E out of the post-proposal game by restricting E's participation to a pre-proposal phase changes the game entirely and is more likely to solidify and increase mistrust and hostility toward the USFS.

Notes

1. As noted earlier, one can also assume that external actors move first in the game by offering a proposal that can be either accepted or rejected by the USFS. Our analysis reveals that the outcomes in the game do not change depending on which actor moves first. Thus, no generality is lost by assuming the USFS moves first in the game.
2. There is no guaranteed payoff to either side when a Fight occurs. The court may rule completely in favor of the USFS or the environmentalists or a compromise outcome may be handed down.
3. Not all environmental groups can be characterized as resolved. Some E groups may prefer the USFS winning to a Fight (an irresolute E), while others may view the Status Quo as the worst possible outcome. An anonymous reviewer called the latter environmentalists, "ideologues," and we are thankful they brought to our attention an E type we had not previously considered.
4. Complete information means that the payoffs are known (there is no uncertainty in the game). To find an equilibrium using backward induction, begin at the end of the tree to determine which act will be selected. Once that is determined, move backward in the tree to figure out what the second to last player will choose and so on until you arrive at beginning of the tree. The following illustrates the method of finding the equilibrium via backward induction. At its last decision node, an irresolute USFS will defend itself in court since a Fight is preferred to the outcome it obtains when it does not defend itself, USFS Loses2. E knows that if it files suit, the USFS will defend and the result will be a Fight. If E does not file, the result is USFS Wins2. Since the former is preferred to the latter, E files suit. The USFS knows if it rejects E's appeal, it will get a Fight. If it accepts the appeal, it will obtain the lesser preferred outcome, USFS Loses1. The USFS rejects the appeal. For E, appealing a decision leads to a Fight which is preferred to the outcome it gets for failing to appeal, USFS Wins1. The USFS knows that if it makes a proposal, the outcome is a Fight, and if it makes no proposal, the outcome is the Status Quo. Since an irresolute USFS prefers the Status Quo to a Fight, the USFS chooses not to make a proposal. The same logic is used to determine the outcome, Fight, when the USFS is resolved. A resolved USFS knows that if it makes a proposal, the outcome is a

Fight, and if it makes no proposal, the outcome is the Status Quo. Since a resolute USFS prefers a Fight to the Status Quo, the USFS chooses to make a proposal.

5. Appeals Reform Act of 1993, 16 U.S.C. 1612 note.
6. The rule change also eliminates the requirement that environmental studies be completed before trees are harvested.
7. Fed Reg., Vol. 68, #107, 4 June 2003.
8. Healthy Forest Restoration Act of 2003, P.L. 108–148.
9. P.L. 108–148, Sec. 106.
10. For example, under the new guidelines judges are expected to consider whether an act that may pose a potential risk to endangered species is potentially more or less harmful than the possibility of a fire (and the possible effect on the species) that may result if no action is taken to reduce hazardous fuels.
11. The USFS is dependent on private timber companies to remove trees and underbrush as part of hazardous fuel reduction actions. Those trees, however, are often dead, dying, or of a small diameter, and therefore have limited economic value. Thus, to pay for fuel reduction activities the USFS must depend on Congressional appropriation or it must provide access to timber with a greater commercial value, such as large, old-growth trees.
12. The USFS also wins if we assume the appeals process is eliminated for E. The outcome is USFSWins2.
13. See Healthy Forest Restoration Act of 2003, P.L. 108–148, and Fed Reg., Vol. 68, #107, 4 June 2003.

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